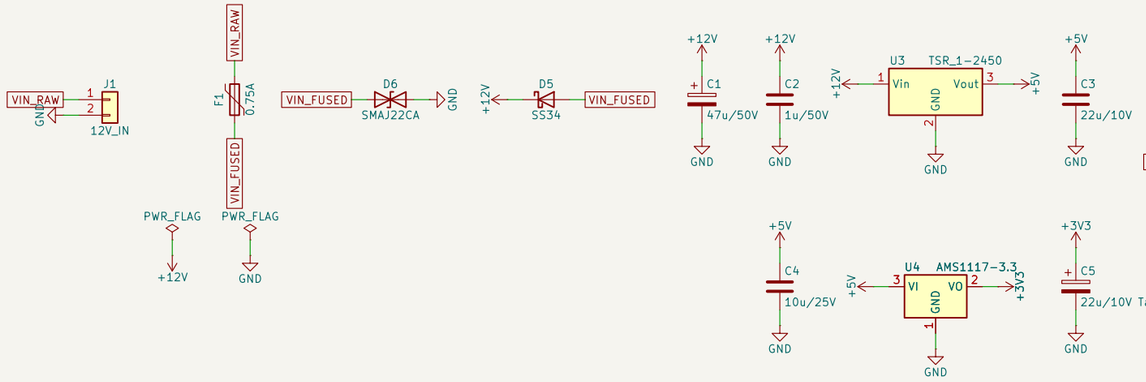
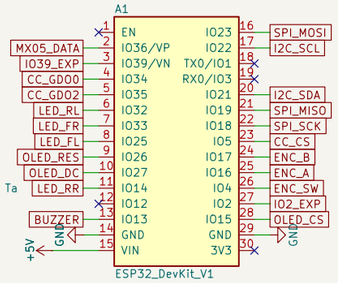
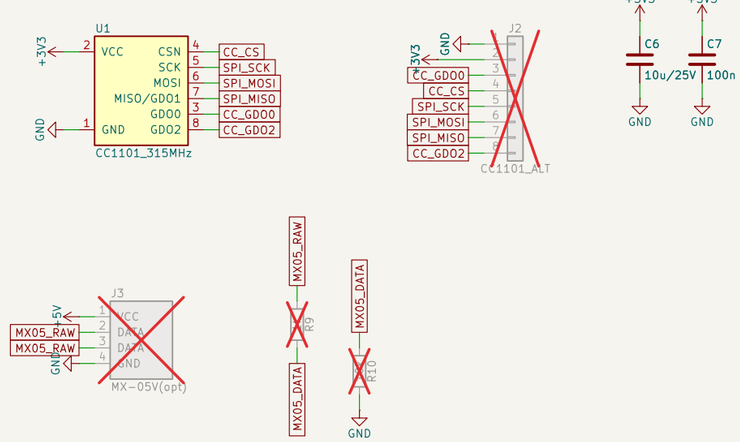
POWER: 12 V (ACC) -> 5 V -> 3.3 V



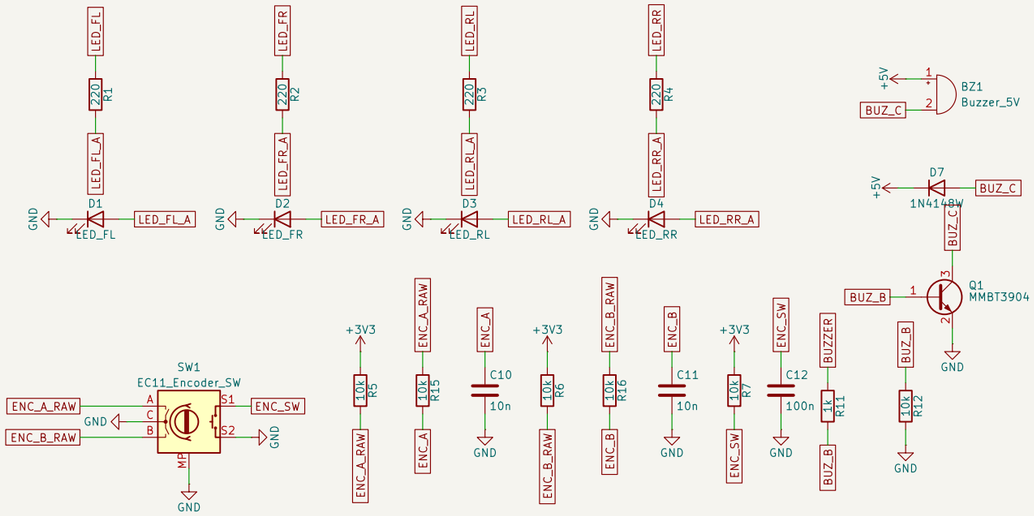
MCU: ESP32 DevKit V1 (DOIT, 30-pin)



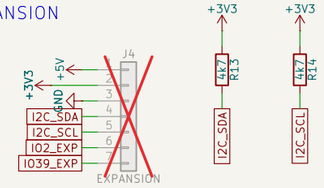
315 MHz RADIO: CC1101 module (+ optional MX-05V)



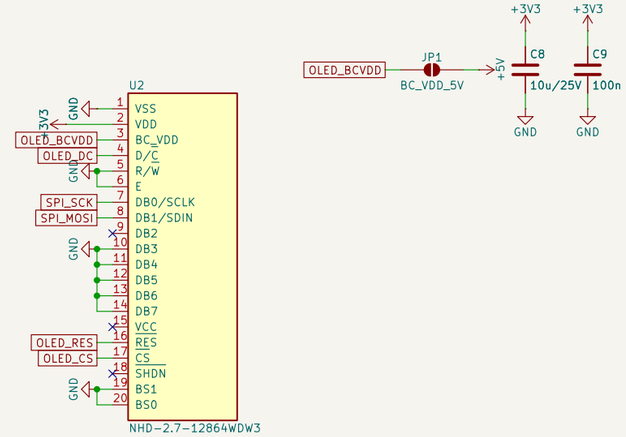
FRONT PANEL: tyre LEDs, menu encoder, buzzer



EXPANSION



DISPLAY: NHD-2.7-12864WDW3 (SSD1322, 4-wire SPI)



ESP32 GPIO MAP
IO18 SPI_SCK -> CC1101 SCK + OLED SCLK
IO23 SPI_MOSI -> CC1101 MOSI + OLED SDIN
IO19 SPI_MISO <- CC1101 MISO
IO5 CC_CS   -> CC1101 CSn
IO34 CC_GDO0 <- CC1101 GDO0 (OOK data)
IO35 CC_GDO2 <- CC1101 GDO2 (carrier sense)
IO15 / IO27 / IO26 -> OLED CS / DC / RES
IO25 / IO33 / IO32 / IO14 -> LED FL/FR/RL/RR
IO16 / IO17 <- encoder A / B (PCNT)   IO4 <- push
IO13 -> buzzer   IO36 <- MX-05V data
IO21 / IO22 I2C SDA / SCL   IO2 + IO39 (input/ADC) spare
Use a WROOM-32 DevKit (on WROVER, IO16/IO17 are taken by PSRAM)

NOTES
1. CC1101 and OLED are 3.3 V parts: never drive them with 5 V logic.
2. Target sensor: GM / Schrader EG53MA4 (FCC MRXEG53MA4): 315 MHz OOK,
   Manchester, ~123 us chips. Run the CC1101 in async serial mode. data on GDO0.
3. U3: 7805-pinout switcher rated >= 36 V in (TSR 1-2450, K7805-1000R3). Not R-78E (28 V max).
4. JP1 open by default. Close only after re-jumpering the NHD module to
   option #1 (R4 open, R5 closed) so its OLED boost runs from 5 V.
5. J3, R9, R10 (MX-05V superregen receiver) optional: DNP with the CC1101.
6. J2 = 1x8 breakout of the CC1101 bus for modules with another pinout.

MECHANICAL



Generated by hardware/scripts/build_tpms_panel.py — edit design.py and re-run
ESP32 DevKit V1 + CC1101 (E07-M1101D) + NHD-2.7-12864WDW3 OLED
Ajaya Dahal

Sheet: /		
File: tpms_panel.kicad_sch		
Title: TPMS Monitor Panel (315 MHz)		
Size: A3	Date: 2026-11-09	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 1/1